

EL CALAFATE LAGO ARGENTINO, Argentina

WMO: 879030

Lat: 50.336S

Lon: 72.249W

Elev: 220

StdP: 98.71

Time Zone: -3.00 (W03)

Period: 91-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	-6.6	-5.3	-13.3	1.2	8.6	-11.7	1.4	7.6	11.9	5.1	10.4	3.7	0.9	N/A	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.9	22.2	11.9	20.7	10.7	19.2	10.0	12.9	20.3	11.8	18.7	11.0	17.6	5.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
9.5	7.6	14.3	8.1	6.9	14.1	6.9	6.3	13.6	37.2	20.7	34.4	18.7	32.2	17.6	19.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.6	12.2	10.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-10.7	16.0	1.4	1.5	-11.8	17.1	-12.6	18.0	-13.4	18.8	-14.4	19.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	13.4	12.8	11.1	8.3	4.5	0.5	0.7	3.1	5.2	8.4	10.9	12.1
	DBStd	5.36	2.65	2.96	2.76	2.96	3.07	3.12	3.58	2.94	3.15	2.90	2.65	2.73
	HDD10.0	1298	3	7	19	67	172	284	289	214	147	65	21	10
	HDD18.3	3938	155	156	223	301	429	534	547	473	395	307	224	193
	CDD10.0	406	107	85	54	17	1	0	0	2	16	47	76	
	CDD18.3	2	1	1	0	0	0	0	0	0	0	0	1	
	CDH23.3	14	4	6	1	0	0	0	0	0	0	1	3	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	3.4	5.2	4.3	2.9	2.4	1.9	1.3	2.2	3.0	3.5	4.4	4.9	5.1
Precipitation	PrecAvg	200	12	9	14	26	30	22	28	20	17	13	9	13
	PrecMax	364	58	32	41	82	75	78	77	47	42	37	36	82
	PrecMin	59	0	0	0	5	0	0	0	0	0	0	0	0
	PrecStd	86	12	8	12	21	21	19	18	13	12	11	9	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.8	24.1	22.5	18.6	14.3	10.2	11.6	12.7	16.2	18.6	21.1	23.6
		MCWB	12.6	13.6	12.6	10.3	7.5	5.2	5.6	5.2	7.3	8.8	10.3	12.7
	2%	DB	21.7	22.0	19.8	16.3	11.8	8.2	9.7	10.9	13.4	16.7	19.2	21.1
		MCWB	11.3	12.2	10.7	9.1	6.3	3.9	4.5	4.9	5.7	7.8	9.6	10.5
	5%	DB	20.2	20.0	17.9	15.0	10.4	6.7	7.9	9.3	11.9	15.2	17.8	19.1
		MCWB	10.3	11.1	9.7	8.3	5.3	2.8	3.3	3.9	5.1	7.0	8.8	9.9
	10%	DB	18.7	18.2	16.3	13.3	9.0	5.1	6.0	7.9	10.5	14.0	16.4	17.6
		MCWB	9.4	9.8	8.9	7.2	4.6	1.8	2.4	3.0	4.4	6.7	8.0	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	14.5	13.3	12.6	8.3	5.7	6.3	6.6	7.6	10.3	13.0	13.8
		MCDB	21.9	21.7	20.5	16.4	12.4	9.3	10.6	11.2	14.4	15.9	18.6	21.4
	2%	WB	12.1	12.8	11.8	9.9	7.0	4.3	4.7	5.2	6.4	8.7	10.7	12.0
		MCDB	19.8	20.4	17.3	15.2	10.5	7.7	8.8	9.7	12.5	14.6	17.1	19.3
	5%	WB	11.1	11.5	10.8	8.7	6.1	3.1	3.7	4.3	5.5	7.9	9.6	10.7
		MCDB	18.8	18.5	16.2	13.6	9.3	6.1	7.2	8.5	11.0	13.8	16.4	17.4
	10%	WB	10.2	10.5	9.8	7.7	5.2	2.1	2.6	3.4	4.6	7.1	8.7	9.6
		MCDB	17.8	17.1	15.3	12.4	8.2	4.7	5.9	7.4	9.7	13.1	15.4	16.1
Mean Daily Temperature Range	5% DB	MDBR	8.9	8.9	8.5	7.8	6.3	5.5	5.7	6.7	7.7	8.8	9.5	9.0
		MCDBR	10.8	10.8	10.5	9.7	8.1	7.2	7.7	8.5	9.8	10.6	11.2	10.8
		MCWBR	5.6	5.8	5.9	5.9	5.1	4.7	5.1	4.9	5.5	5.8	6.4	5.7
	5% WB	MCDBR	9.9	9.9	9.1	8.7	7.4	6.8	7.4	8.1	9.3	9.5	10.3	9.4
		MCWBR	5.7	5.9	5.7	6.3	5.2	4.8	5.2	5.3	5.6	5.9	6.8	6.1
Clear-Sky Solar Irradiance	taub	0.308	0.306	0.293	0.281	0.259	0.245	0.249	0.266	0.287	0.299	0.300	0.310	
	taud	2.553	2.569	2.599	2.595	2.525	2.397	2.385	2.419	2.463	2.517	2.542	2.527	
	Ebn at Noon	1004	974	933	854	773	717	761	847	918	970	1007	1010	
	Edh at Noon	101	93	79	64	51	49	56	73	88	97	102	106	
All-Sky Solar Radiation	RadAvg	5.83	4.91	3.40	2.06	1.18	0.83	1.02	1.75	2.98	4.25	5.47	5.97	
	RadStd	0.37	0.28	0.19	0.13	0.11	0.07	0.09	0.18	0.23	0.25	0.30	0.37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page